Business insights using knowledge graphs by text analytics in dynamic environments

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ABSTRACT

Business Intelligence (BI) requires the collection and organization of important pieces of information (i.e. entities) from multiple sources to provide valuable insights (e.g. business trends) as events (i.e. a specific happening linked with a specific location and time) to users. Online news articles are one of the important information sources that present business news offered by various companies in the market every day all around the world. These news articles often cover the same events and report redundant information. Existing news platforms aim at collecting the key entities from news articles and providing a mechanism to view the latest and relevant business events based on user interest. However, they do not provide a method to model business events and understand them temporally, spatially, and contextually (i.e. changes in the event). For instance, it is crucial to know for how long a business event has been active? How important is its evolution locally, or worldwide? Or how did different companies come up with this event as competitors in the market? The contribution of this research is the exploration of the possibilities of modeling spatial, temporal, and contextual information evolution related to business events through the application of knowledge graphs and text analytics, more specifically, Natural Language Processing (NLP) methods. The constructed knowledge graphs through Named-Entity Recognition (NER), i.e., an NLP technique, present a compact news representation that tells the key entities of the business event at one glance using linked open data concepts. It enables the assessment of other related news events as well as provides the means for analysis of the influence and evolution of business events.

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CCS CONCEPTS

• Artificial Intelligence • Information Systems Applications • Document Management and Text Processing

KEYWORDS

Artificial Intelligence (AI), Business Intelligence (BI), Natural Language Processing (NLP), Linked Open Data, Knowledge Graphs

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1 INTRODUCTION

Online news articles are published by numerous news agencies every day [1]. The market changes and company movements are reported in news articles provide the important business information [1]. Identifying and collecting different business entities from various news articles (e.g. public tenders, cadastral records, etc.) across borders and in different languages for constructing business events is a tedious and expensive task [2]. Here, the term business event refers to the activity (i.e. investing, marketing, developing, etc.) performed by a company at a specific time and a specific location [1, 2]. Business events retrieved from the news articles are fused with the company's (i.e. creator of events) information [2] and are immediately available to business analysts who are interested in analyzing their evolution in the dynamic market. Public bodies are publishing open data about firmographics and contextual databases of companies to ensure the availability of company data to users [2]. The basic information about companies such as names, registered addresses, ownerships, related products, and services, etc. is the foundation that many business analysts depend upon for analyzing business events [2].

Recently, there are numerous attempts made based on advanced techniques of machine learning combined with text analytics (i.e. deriving insights from text) for extracting real-time information from news articles and formulating business

events [1 – 3]. However, the motivation of this article is not to present a novel event extraction method but to construct a data model using knowledge graphs (i.e. a directed, labeled, multi-relational graph with some form of semantics) [4] based on linked data concepts (i.e. available data in machine-readable format) [5] to understand how business news evolves. Although, a brief overview of the state-of-the-art techniques on event detection [6] is presented. Later, created knowledge graphs are populated with the information entities extracted through a basic NER technique [7] for representing business events. Understanding the evolution of business events is useful for predicting upcoming business market trends. For instance, in predicting whether a business event is disappearing by going relatively unnoticed, or intensifying and reverberating the creation of other similar business events [8]. This is of high interest to business analysts. When investigating a business event, the most important aspects are the contextual information about the company and the spatio-temporal evolution, i.e. how business events change over time over different geographical regions. In the domain of BI, several models (e.g. ontologies) are presented in the existing literature [1, 2, 8, 9] by combining information about events extracted from news articles through NLP techniques with the companies' information for constructing business events. There are few attempts in the literature [9] to link similar news articles together based on temporal expressions that occur in the articles. However, maintaining the spatial as well as contextual evolution of business events is not yet explored in the literature. Furthermore, the majority of the presented approaches are not aligned with semantic web technologies, linked data, and the W3C technology stack aimed at forming a global interconnected data graph [10]. To address this knowledge gap, this work presents a data model which can hold the information of static and changing business entities (i.e. business events, company data, and geographical locations in our case) over time [11] by exploiting the concepts of linked data. Moreover, the enrichment of the constructed knowledge graph with data is achieved using the NER technique.

The paper is structured as follows. After a literature review of related work in the following section, i.e. Section 2, we have presented our data model for news articles along with the instantiation example in Section 3. In Section 4, we discussed our system and demonstrated its functionality. Section 5 is about the conclusion and the future work.

2 BACKGROUND

Event extraction from news articles has emerged as one of the tasks of Information Extraction (IE) (i.e. the process of extracting information from text) which is an area of NLP [12]. In the literature, there are primarily three approaches to event extraction [6], which are; 1) data-driven approaches, 2) expert knowledge-driven approaches, and 3) hybrid event extraction approaches that combine knowledge and data-driven approaches. Data-driven approaches are used for NLP applications and are based solely on quantitative techniques for discovering relations [6]. They need large text corpora for developing models for approximating linguistic phenomena. Using these models, the data is transformed into knowledge through the exploitation of statistics, machine learning, and linear algebra [6]. These approaches don't use linguistic resources or domain knowledge. Several attempts using data-driven approaches are found in the literature. A few of them are [13, 14]. The drawback of these approaches is that they do not incorporate the meaning explicitly of entities for constructing events [6]. They use corpora for discovering relations without considering the semantics. Also, a large amount of data is needed for achieving statistically significant results. In contrast to data-driven methods, knowledge-driven text mining uses predefined linguistic patterns for expressing rules to represent expert knowledge. These methods rely on linguistic, lexicographic knowledge, and domain knowledge. Various attempts are present using knowledge-driven techniques, such as [15, 16]. There are benefits of using knowledge-driven approaches such as knowledge-based approaches that use less training data than data-driven approaches. Also, lexical, syntactical, and semantic elements are combined for defining useful patterns for knowledge extraction [6]. However, knowledge-based approaches get complicated when the patterns need to be scaled up to cater to more situations, as the patterns are hand-tuned. Data-driven and knowledge-driven approaches present their advantages and disadvantages. Merging these two approaches could give us the best results [17]. However, in hybrid event extraction systems, the amount of required data and the level of expertise increases due to the merging of multiple techniques [6].

Once events are extracted, it's important to link them to establish understanding and their evolution [18, 19]. Event linking provides business analysts a context and background about the event they are analyzing at present [20]. Linking news events is addressed using a variety of directions [9, 18 – 20]. Most common approaches are based on topic detection and tracking using similarity metrics such as Term Frequency–Inverse Document Frequency (TF-IDF) and Latent Dirichlet Allocation (LDA) methods. For instance, [21] used collective factorization for modeling the temporal evolution of news. In addition, [9] used temporal expressions occurring in the news articles and proposed the arrangement of these articles in chronological fashion based on their relatedness. [18] introduced event linking using a past event reference in context, enables searching of the articles in a news archive, and canonically labels an event reference with the article where it was first published. Moreover, [19] presented an approach for cross-linking heterogeneous news events. It used dynamic topic modeling for finding the events and tracking

their evolution over time for understanding the temporal publishing patterns. Furthermore, [20] proposed a model using LDA method for retrieving other news articles that provide important context related to the news article that is being looked into.

The above-mentioned studies discuss the understanding of news events over time. However, the current body of literature is missing the data modeling based on contextual as well as spatial evolution of news events. Though, research by [22] has proposed a model for representing events using a semantic network and linking them based on contextual and temporal information for supporting event-centric applications. However, their model is theoretical. The application details of the methods, platforms as well as the state-of-the-art ontologies which can be exploited to construct knowledge graphs are lacking. Therefore, the goal of this paper is to present a model that is aligned with semantic web technologies, linked data, and the W3C technology stack aimed at forming a global interconnected data graph for understanding the spatial, temporal, and semantic evolution of various news events over time. The application of news events for this article has kept focused on the business domain.

3 PROPOSED SYSTEM

3.1 Data Model Construction

The proposed system contains 5 key steps (see Fig. 1) to explore business insights using the events extracted from news articles. The 1st step is to collect the business news articles from their websites. The process of web crawling is used to scrape the web for retrieving news articles. The retrieved textual data files of news articles are pre-processed. Basic pre-processing steps such as stop-word removal, filtering of special characters, and hashtags is executed [10]. Then, lemmatization is performed to lemmatize the words.

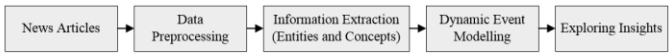

Figure 1: Steps involved in exploring business insights.

To capture the important entities for representing business events from the collected news articles, the fundamental requirements were initially identified through the literature review. These requirements were primarily based on 3 different types of information that evolves over time (see Fig. 2) about:

1) Business event in a news article (e.g. the title, description, location, homepage, domain, a person of interest, writer, and the publisher),
2) The company that has published a news article (e.g. company title, address, number of employees, contact email, homepage, type of industry, the last turnover, products, and services of the company), and
3) The spatial (e.g. city, region, etc.) details which are mentioned in the article.

Figure 2: Entities of a dynamic business environment.

Each information entity mentioned in Fig. 2 evolves upon the occurrence of any change in them. A change can occur in the location, contextual attributes of a company, or in a news event. These changes can occur independently or concurrently. These information requirements were later transformed into a set of 19 competency questions (see Table 1). The requirement questions are categorized as contextual, spatial and temporal based on their nature of the information. Moreover, related existing ontologies and controlled vocabularies were studied to model the information for addressing the competency questions. A set of classes and properties from state-of-the-art controlled vocabularies are selected (see Table 1) for defining the concepts and their relationships, which are; 1) Resource Description Framework Schema (RDFS), DCTERMS, DCAT ontology, and vCard ontology. In addition, the model uses GeoNames for specifying the spatial features and OWL-Time ontology for describing temporal concepts. The graphical representations of the above-mentioned concepts are shown in Fig. 3 and 4.

3.2 Incorporating the Dynamicity of Events over Time

To model the dynamicity (i.e. in terms of change in the location, a company or an event details) of business events over time, 4D-fluent, a state-of-the-art approach [11] is used to show how temporal information can be represented in Web Ontology Language (OWL) (i.e. a semantic web language to represent knowledge about entities, and relations between entities [23]. The basic idea is that each entity has a start and

an end point. An entity is seen as a four dimensional having the temporal part [11]. Changes occurring on the properties of the temporal part keeping the entity as a whole unchanged. To add the time dimension to a model, classes Timeslice and TimeInterval with properties TimesliceOf and tsTimeInterval are used [11]. The class TimeSlice is used for representing temporal parts (i.e. timeslices) of entities and class TimeInterval is the domain class of time intervals. A TimeInterval holds the temporal information of a TimeSlice. Property tsTimeSliceOf links an instance of class TimeSlice with an entity, and property tsTimeInterval links an instance of class TimeSlice with an instance of class TimeInterval. The properties with a time dimension are the Fluent properties and links instances of class TimeSlice. Based on this approach, the model is built (see Fig. 5).

Table 1. Data requirements for capturing the information about business event.

No.	Requirement questions	Type	Entity	Ontologies and vocabularies used
1	What is the name of a company?	Contextual	Title	dct:Title
2	What was the last turnover of a company?	Contextual	Turnover	rdfs:Literal
3	What is the address of a company?	Spatial	Address	vcard:hasAddress
4	What are the number of employees in a company?	Contextual	Employees	rdfs:Literal
5	What is the contact email of a company?	Contextual	Email	vcard:hasEmail
6	What is the official website of a company?	Contextual	URL	dcat:AccessURL
7	What products does a company produce?	Contextual	Keyword	dcat:Keyword
8	What services does a company offer?	Contextual	Keyword	dcat:Keyword
9	Which industry a company belongs to?	Contextual	Keyword	dcat:Keyword
10	What is the title of a news article?	Contextual	Title	dct:Title
11	What is the description of a news article?	Contextual	Description	dct:Description
12	Who is the writer of a news article?	Contextual	Writer	dct:Contributor
13	Who is the person of interest in a new article?	Contextual	Person	dct:Subject
14	What is the domain of a news article?	Contextual	Domain	dcat:Keyword
15	What is the location entity mentioned in a news article?	Spatial	Location	gn:Feature
16	What is the company name mentioned in a news article?	Contextual	Name	rdfs:Label
17	Who is the publisher of a news article?	Contextual	Publisher	dct:Publisher
18	Where to access full news article?	Contextual	URL	dct:DownloadURL
19	What is time entity as a deadline found in a news article?	Temporal	Date	dct:Temporal

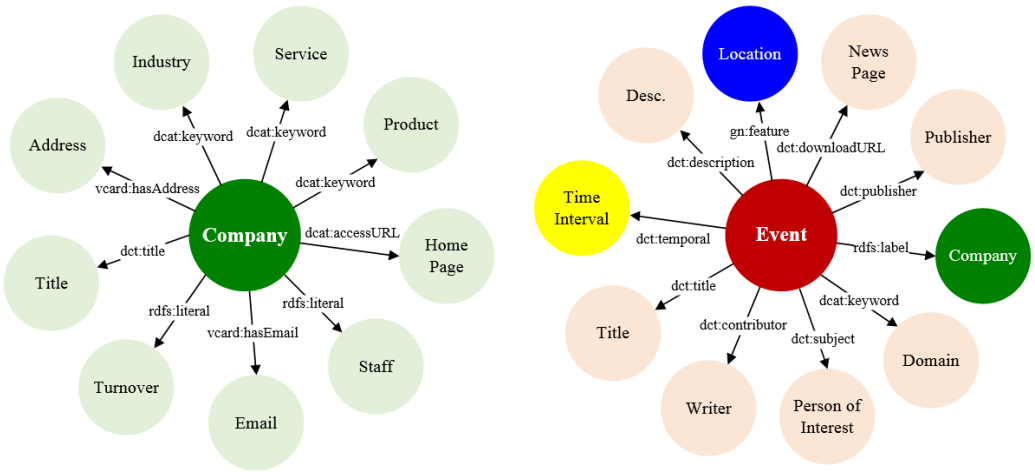

Figure 3: Entities concerning the company details are shown on the left and entities concerning the business events are shown on the right.

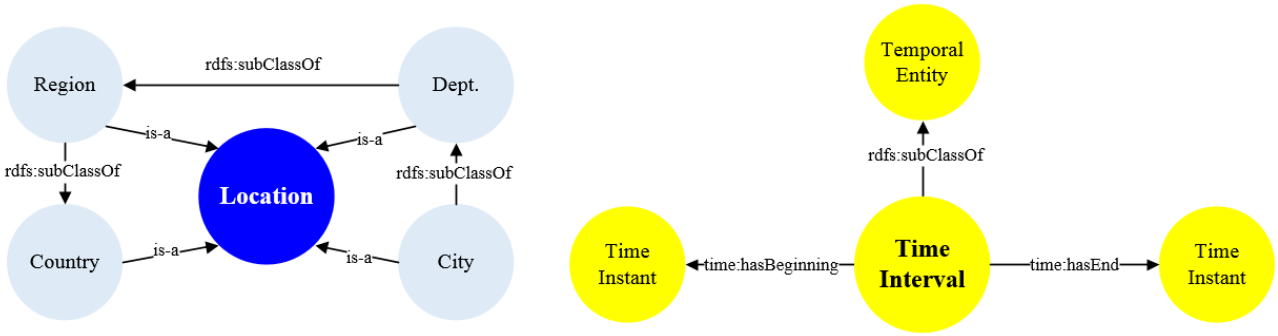

Figure 4: Entities concerning the location are shown on the left and entities concerning the time are shown on the right.

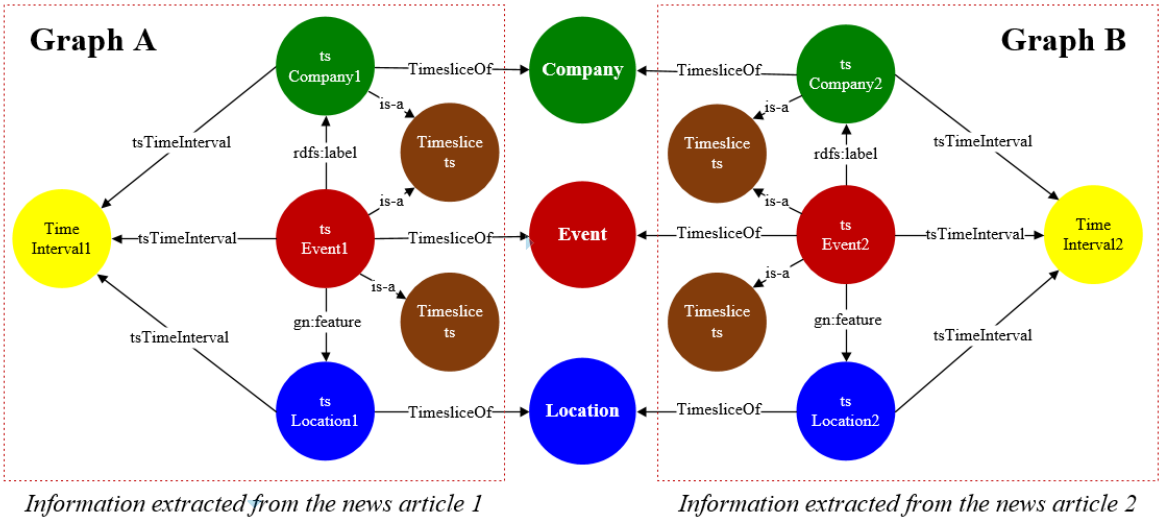

Figure 5: Dynamic knowledge graphs having the properties of a company, event and a location over different time

3.3 Instantiation of the Model

Based on the knowledge graph entities presented in Fig. 2, there can be several scenarios where the graphs are holding the information of static and dynamic business entities. However, to show the proof-of-concept application, an example is presented using the extraction of information entities from the news article for populating the knowledge graphs. To extract information entities (i.e. named-entities), the SpaCy library is used for processing texts of news articles [10]. For the 1st news article, Herige, a company, published its article on 3rd May 2022 (see Fig. 6) about the acquisition of the concrete subsidiary to expand its business in the region of Aquitaine [24]. After two weeks, the same company 'Herige' published its news article [25] to allocate more resources to expand its business (see Fig. 7).

Both the news events are from the same company i.e. Herige' based at Aquitaine region of France. Also, the possible domains of both the news articles which can be inferred from the text are related to concrete industry and building material. Whereas, the nature of business events is different in terms of the contextual information. The 1st news article illustrates the acquisition of the concrete subsidiary. Whereas, 2nd article addresses the allocation of resources to increase the business. In addition, there is no spatial location (i.e. location) mentioned in the 2nd news article. To model these two news events, Fig. 8 illustrates a graph with classes Company with datatype property companyName, Event with datatype property eventTitle, and Location with datatype property locationName. In this example, the eventTitle and locationName are dynamic (fluent) properties (i.e. their values may change in time). We kept the location dynamic as the location is not defined in the 2nd news article. Whereas, the companyName is the static property (i.e. its value does not

change in time). Using this approach, two similar events will be linked together (see Fig. 9)

Herige ORG , an acquisition in Aquitaine NORP
Herige announces the acquisition by its concrete subsidiary Edycem ORG , of the family company Audoin et Fils Béton ORG based in the Aquitaine NORP region, providing the group with new high-performance power plants, included in the consolidated accounts on 1 May. Composed of three concrete plants, a platform bringing together the trading of aggregates, the treatment of inert and an aggregate washing installation, the perimeter employs 31 employees. 'Thus, Herige ORG completes its network by creating a stronghold in the Aquitaine basin and continues its dynamic of innovation by developing actions in favor of the circular economy, in particular through the production of recycled aggregates', explained.

Figure 6: New article published on 3rd May 2022 of the company 'Herige' about expanding its business in Aquitaine region of France.

Four things to know about Herige ORG 's transformation plan: The Herige ORG group focused on the world of construction through the trading of materials, the concrete industry and industrial carpentry has just communicated a new roadmap to accelerate its growth. Thus, the company intends to allocate an envelope of €50 million MONEY to accelerate financing on these themes: "This is a normative investment rate. What changes is its distribution: we have chosen to allocate more resources and transformation topics", says Benoît Hennaut PERSON , the Chairman. A first phase will consist in particular of modernizing low-carbon silos, fleets of trucks or vehicles or even forklifts. To finance these investments, Herige ORG plans to seek more green loans. The group also intends to rely on the company Audoin, to "deploy new recycling platforms in our regional network, probably from the end of 2022. This will be either in the form of mobile units that can be installed on large construction sites, or in the form of fixed power plants. We are still in the costing phase", announces Benoît Hennaut PERSON .

Figure 7: New article published on 17th May 2022 of the company 'Herige' about allocation of more resources for expanding its business.

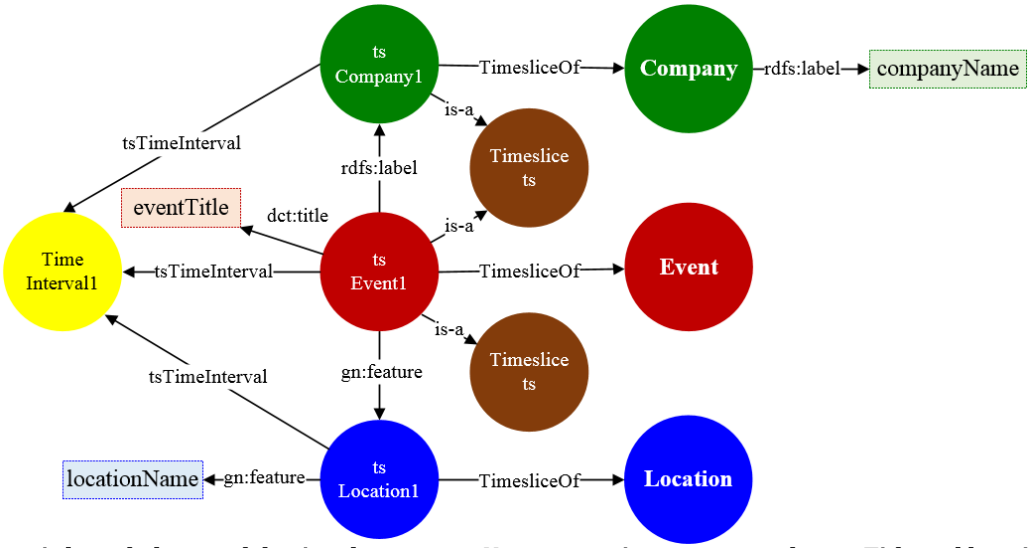

Figure 8: Dynamic knowledge graph having the companyName as static property, and eventTitle and locationName as dynamic properties.

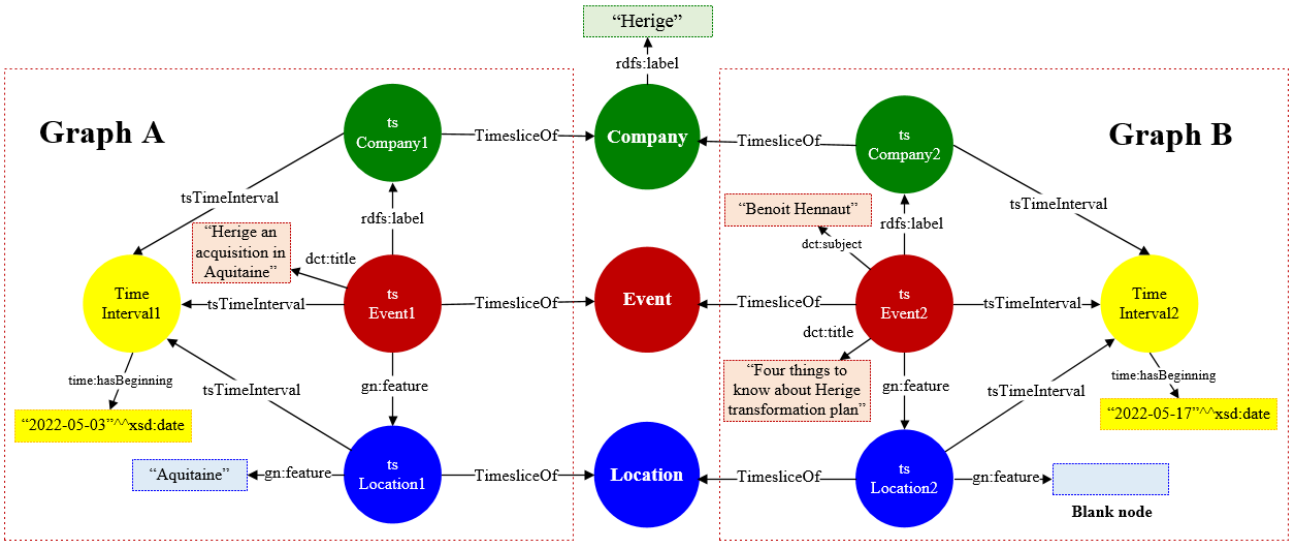

Figure 9: Dynamic knowledge graphs modeling with an instantiation example.

3.4 Implementation

Once the data modeling through knowledge graphs is achieved and information entities are extracted for the representation of business events from the news articles, the knowledge graphs are saved in a database. For storing the graphs, RDFLib, a Python library, is selected for working with RDF data. The RDF platform is chosen as it is built on the existing Web standard and enables effective data integration from various knowledge sources. This allows several ontologies and vocabularies to be applied, linked, and queried without modifying the data instances. RDFLib has a graph interface to parse, store, query, and serialize RDF triples (i.e. consists of a subject, predicate, and object) [26]. In our case, 'event' is a subject, 'dct:title' is a predicate, and 'title' (i.e. a literal) is an object (see Fig. 2). The queries can be made on RDF graphs using triple patterns or SPARQL (i.e. a standard query protocol for linked open data and RDF databases) queries. The knowledge graphs representing similar business events need to be kept together along with their spatio-temporal evolution.

For this, first, we need to define the criteria for detecting the similarity of events, and second, to answer how much two similar events should be linked together. To check the similarity of events, the entities concerning the 'Title' and 'Domain concepts' of knowledge graphs are considered (see Fig. 1) and compared using the Cosine similarity index [27]. However, the criteria of selecting the entities can be changed. Cosine similarity is a measure of similarity between two concepts [27]. For its calculation in terms of a score interval (-1,1), the concepts are viewed as vectors. The vectors of concepts are extracted using the pre-trained word vectors' dataset [28]. Based on the Cosine similarity scores, similar knowledge graphs are organized together. Similar graphs are stored as subgraphs under the same 'ConjunctiveGraph', i.e. a top-level representation of a graph in RDFLib. ConjunctiveGraphs enable the aggregation of all the contexts (subgraphs) within it and are distinguished by unique identifiers. These identifiers can be assigned automatically by the RDFLib or can be supplied at the creation time of a graph. The value of the identifier can be an RDFLib Literal. For instance, in our case, to generalize two news articles (as discussed above), the identifier for their ConjunctiveGraph can be a set of domain concepts i.e. represented as a Literal, e.g. "Acquisition, Building material, and Transformation". Using this method, knowledge graphs representing similar business events are saved together.

4 DISCUSSION AND CONCLUSION

The article presents the modeling of spatial, temporal, and contextual information evolution of business events through the application of knowledge graphs. For this, first, the fundamental requirements for understanding business events were identified. These requirements were mainly based on 3 types of information that evolve, which are; 1) something that happened (i.e. an event), (2) the place where it happened (i.e. location), and (3) the entity with which it was associated (i.e. company). In addition, information concerning who was involved in the event (a person of interest) is also considered. Moreover, to include the profile of the company that is advertising the information, the information related to its last turnover, staff, and its industry is also modeled. These pieces of information are crucial for understanding the company's resources, the type of resources, and its position in the

business market. Later, related ontologies and controlled vocabularies were selected for creating the knowledge model. The proof-of-concept implementation of the data model is executed using the Python library, i.e. RDFLib for constructing knowledge graphs. The reason behind selecting this library is because it is open-sourced, and fairly simple for manipulating RDF data with minimal technical expertise [26]. Once the graphs are generated, it is easy to populate with relevant named entities and link them based on their similarity to maintain the spatial, temporal, and thematic evolution.

The novelty of this work is in combining techniques of knowledge graphs and NLP techniques for providing an informative and engaging view of business events spatially, temporally, and contextually over time. The spatial evolution of an event can reveal whether an event is local or has a particular global economic or political facet. Also, the temporal evolution will indicate the business news popularity, significance, and duration. Moreover, the contextual evolution of events will show how different companies come up with the same event as com-petitors in the market. Ultimately, the relation between the spatial, temporal, and contextual evolution of business events may also reveal valuable insights. The main limitation of the work is that knowledge graphs are populated manually. However, with the increasing amount of business news articles and the exploding number of digital data sources, it becomes crucial to keep the processes of build-ing knowledge graphs for constructing business events automatic and real-time. But, to achieve this, it should be noted that the amount of expertise needed is high and the accuracy of finding the accurate information entities gets complex in real-time processing. Future work can be done to automate the processes using advanced machine learning techniques.

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